

Research Proposal: Logic, Design, and Delivery

AIHA8121: Research Methodology & Ethics

Week	Content Coverage	Date	
1	Introduction to Research Methodology and Ethics	2026/9/5	2026/9/5
2	Literature Review	2026/9/12	2026/9/18下午
3	Learning LaTeX & Writing Research Proposals I	2026/9/19	2026/9/19
4	Writing Research Proposals II	2026/9/26	2026/10/10
5	Writing Research Papers I	2026/10/3	2026/10/10晚上
6	Writing Research Papers II	2026/10/10	2026/10/17
7	Writing Research Papers III	2026/10/17	2026/10/17晚上
8	Research Ethics	2026/10/24	2026/10/24
9	Making Posters and Slides	2026/10/31	2026/10/31
10	Making Posters and Slides	2026/11/7	2026/11/7
11	Making Posters and Slides	2026/11/14	2026/11/14
12	Individual Student Presentation I	2026/11/21	2026/11/21
13	Individual Student Presentation II	2026/11/28	2026/11/28
14	Individual student presentation III	2026/12/5	2026/12/5
15	Individual student presentation IV	2026/12/12	2026/12/12

Proposal = Argument + Plan

WHY	WHAT	HOW	WHO + WHERE
Why this problem matters now	What new knowledge is sought	How credible evidence will be produced	Why the people and setting can deliver

Persuasion comes from evidence and alignment—not from promotional language.

The proposal must make a reasoned case and show an executable route to an answer

CONTENTS

01 Build the case

02 Design the study

03 Feasibility and delivery

What You Will Be Able to Do

01

Diagnose

Find a broken link in a proposal argument

02

Align

Connect questions, data, methods, and analysis

03

Evaluate

Judge feasibility, ethics, risk, and contribution

ONE PROPOSAL = ONE AUDITABLE CHAIN

A strong proposal lets a reviewer follow the reasoning without filling in missing steps

SECTION 01

Build the Case

Move from a worthwhile problem to a bounded, evidence-supported research claim

NEED → GAP → QUESTION → CONTRIBUTION

One Logic, Different Documents

Document	Primary decision	Extra emphasis
Thesis proposal	Is the study suitable and feasible for the degree?	Training value, supervision, scope
Grant proposal	Should limited resources be invested here?	Impact, team, budget, deliverables
Fellowship / position	Can the applicant grow and deliver in this setting?	Trajectory, fit, mentoring

This lecture uses the common core of academic research proposals; the call or institutional template always takes priority.

Keep the core research logic stable, then adapt emphasis to the decision being made

What Reviewers Need to Believe

VALUE

The question is worth answering

Evidence of need and significance

CONTRIBUTION

The answer will add something identifiable

A bounded, documented gap

VALIDITY

The design can support the intended claim

Fit among data, measures, and analysis

FEASIBILITY

The work can be completed responsibly

Access, skills, time, ethics, and fallback

Every section should help the reviewer answer at least one of these four questions

The Proposal Evidence Chain

1 · NEED
Why care?

2 · GAP
What is unresolved?

3 · QUESTION
What will be answered?

4 · OBJECTIVES
What knowledge is sought?

5 · METHOD
How will evidence be produced?

6 · EVIDENCE
What result supports the claim?

7 · CONTRIBUTION
What changes?

If one link changes, revisit the links on both sides.

A proposal is coherent when each link constrains and justifies the next one

Map Any Template to the Same Logic

Template heading	Underlying question	Evidence to show
Background / rationale	Why this problem and why now?	Current evidence, consequence, gap
Aims / objectives	What exactly will be learned?	Specific, bounded objectives
Methodology	How can the claim be tested?	Design, sample, measures, analysis
Work plan / resources	Can it be completed here?	Milestones, access, roles, budget
Outputs / impact	What will be delivered or change?	Outputs, users, indicators, assumptions

Never copy headings without checking what decision each heading supports.

Different headings can serve the same proposal logic, but the required evidence may differ

Write the Summary Last

1

NEED
The consequential problem

2

GAP
What current evidence cannot answer

3

QUESTION
The bounded question

4

APPROACH
Design, data, and analysis

5

FEASIBILITY
Access, team, and time

6

CONTRIBUTION
Expected knowledge or use

Draft the summary after the full logic is stable.

A proposal summary previews the reasoning; it does not report results that do not yet exist

Choose at the Intersection

PREPARATION
Relevant knowledge, methods, and supervision

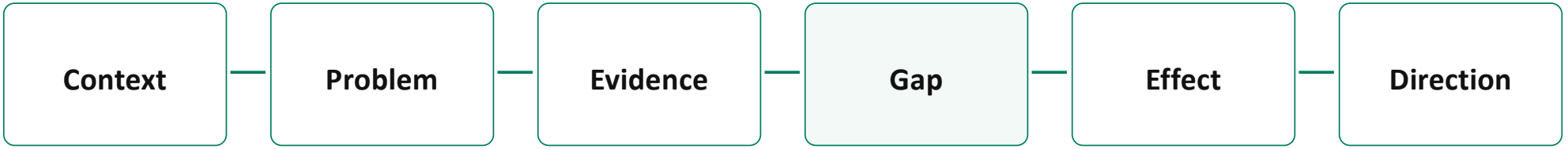
EVIDENCE GAP
A documented unresolved question

ACCESS
Data, participants, equipment, time, and ethics

A valuable, bounded, feasible research question

Originality can come from the question, data, method, setting, comparison, or their combination

Rationale: Context to Consequence



Use evidence to narrow the story; do not make the historical background broader than the decision requires.

A rationale should narrow toward the research question, not expand into a general history of the field

Exercise 1 · Which Rationale Is Stronger?

PAIR TASK

Choose the stronger rationale. Mark the links you can find, then name the missing links.

VERSION A

Hospital readmission is an important problem. Many researchers have developed prediction models. However, few studies have investigated this topic. We will develop a novel deep-learning model.

VERSION B

Readmission models are often tested only at their development sites. Temporal and external performance remains uncertain. We will compare a locked EHR model with a prespecified logistic-regression baseline in later-period and external cohorts.

Illustrative teaching case. Evidence claims require citations in a real proposal.

Begin by identifying the logic; do not rewrite the paragraph yet

Sample Answer: B--Stronger, but Still Incomplete



WHAT VERSION B MAKES VISIBLE

- Evidence boundary: prior evaluation is limited to development sites
- Gap: temporal and external performance remains uncertain
- Planned comparison: a locked model versus a prespecified baseline
- Test settings: later-period and external cohorts

WHAT REMAINS MISSING OR WEAK

- Version B lacks an explicit NEED explaining why the estimate matters
- 'We will compare ...' is a plan, not a research QUESTION
- Version A relies on unsupported 'important', 'few', and 'novel'
- Both versions still require citations and precise definitions

A stronger rationale narrows the claim and makes the next design decision visible

Synthesize, Do Not List

STUDY-BY-STUDY LIST

Study A found…

Study B found…

Study C used…

The reader must infer the pattern.

SYNTHESIS BY QUESTION

What is agreed?

Where do explanations differ?

Which methods drive the disagreement?

What remains uncertain?

The literature review builds an argument about the state of knowledge and the boundary of the gap

Bound and Evidence the Gap

THEORETICAL

Competing explanations are unresolved

Show where predictions differ

EMPIRICAL

A claim lacks adequate evidence

Show limits in data or replication

METHOD

Current measurement or analysis is insufficient

Show the consequence for validity

SETTING / POPULATION

Evidence does not transfer to a relevant context

Justify why transfer cannot be assumed

A gap is not ‘few studies exist’; it is a specific limitation that matters for the proposed claim

Significance: Who Benefits?

SCHOLARLY	PRACTICAL	SOCIETAL
Clarifies a mechanism, estimate, method, or boundary	Improves a decision, workflow, tool, or service when applicable	Supports a broader outcome only when a plausible pathway exists

Not every foundational study must promise policy change or economic value.

Match the claimed significance to the evidence and to the realistic pathway from result to use

Rationale Quality Check

- the problem is specific and supported by current evidence
- the review synthesizes patterns, disagreements, and limitations
- the gap is bounded and consequential
- the proposed question follows directly from the gap
- the significance is proportional to the evidence
- **Report the missing link—not a rewritten title.**

A strong rationale ends with a question that feels necessary, bounded, and answerable

Exercise 2 · Complete the Four-Sentence Scaffold

ILLUSTRATIVE EVIDENCE

- 30-day risk estimates guide limited follow-up resources
- mostly single-institution retrospective development
- limited temporal/external testing

INDIVIDUAL TASK
Write one sentence per line from the evidence above. Do not add unsupported claims.

1 · NEED

The problem matters because ...

2 · EVIDENCE

Existing studies show ...

3 · GAP

However, current evidence cannot determine ... because ...

4 · QUESTION

Therefore, this study asks whether ...

A sentence scaffold reduces the task without supplying the argument for the student

Sample Answer

1 · NEED	Hospitals need reliable estimates of 30-day readmission risk to prioritize follow-up.
2 · CURRENT EVIDENCE	Existing EHR models have often been developed retrospectively within a single institution.
3 · GAP	It remains unclear whether a locked model retains discrimination and calibration over time and at a different hospital.
4 · QUESTION	How do the locked model's discrimination and calibration compare with the prespecified baseline temporally, and are they retained externally?

Each sentence has one job, and the final question follows directly from the gap

SECTION 02

Design the Study

Translate the question into data, measurement, analysis, ethics, and risk controls

QUESTION → DATA → MEASURE → ANALYZE → CHECK

Research Question ≠ Objective

RESEARCH QUESTION

The uncertainty to be resolved

Can an EHR-based model outperform a prespecified logistic regression baseline for 30-day readmission and retain performance at an external site?

RESEARCH OBJECTIVE

The knowledge-producing aim

Develop the model, validate it on a later period, and evaluate the locked model at an external site against the prespecified baseline.

The objective is not simply ‘collect data’ or ‘train a model.’

Questions state the uncertainty; objectives state the knowledge the project will produce

Separate Aims, Work, and Outputs

Element	It answers…	Running example
Objective	What will be learned?	Develop, temporally validate, and externally evaluate the model against the baseline
Activity	What will the team do?	Extract data, train models, run validation
Output	What will be delivered?	Locked model, report, code, or manuscript
Indicator	How is completion observed?	Prespecified metrics reported with uncertainty

One objective may require several activities; outputs can serve more than one objective.

Do not use activities or promised publications as substitutes for research objectives

Four Different Problems

RESEARCH QUESTION
Knowledge uncertainty
Example: Can the EHR model beat the baseline and transfer?

METHOD CHALLENGE
Threat to valid measurement or inference
Example: Class imbalance and missingness

CRITICAL DEPENDENCY
Condition without which work stops
Example: Access to time-stamped EHR extracts

RISK
Uncertain event requiring a response
Example: External site withdraws

Naming the type of problem determines whether you answer it, design around it, secure it, or mitigate it

Align Objectives, Data, and Analysis

Objective	Data	Sample	Measure	Analysis	Validity check
Develop + temporal validation	Time-indexed EHR	Eligible admissions	Outcome + predictors	Nested tuning / CV	Later-period split
External validation	External EHR	Same criteria	Same definitions	Locked evaluation	Calibration + subgroups

If a row cannot be completed, the objective or method is not yet specified well enough

Design, Sample, and Measurement

DESIGN	SAMPLE	MEASUREMENT
•Name the design	•Source population	•Operational definitions
•Justify why it fits	•Eligibility and recruitment	•Reliability and validity
•State comparison and timing	•Size and missingness	•Who measures, when, and how

Include details that affect validity or feasibility; move routine implementation detail to a protocol or appendix.

Method detail is useful when it changes what can be concluded or whether the study can be completed

Analysis Must Match the Estimand

1	**TARGET**	What quantity or comparison is being estimated?
2	**DATA SPLIT**	What data inform development, tuning, and final evaluation?
3	**MODEL + METRICS**	Which analysis and metrics answer the objective?
4	**UNCERTAINTY**	How will confidence intervals, sensitivity, or robustness be shown?
5	**FAILURES**	How are missing data, multiplicity, and assumption violations handled?

Naming the estimand and evaluation boundary prevents the analysis plan from becoming a list of tools

Ethics and Data Shape the Design

ETHICS

- consent or lawful basis
- risk–benefit and vulnerable groups
- fair inclusion and subgroup harm
- independent review and protocol deviations

DATA MANAGEMENT

- data minimization and provenance
- access control, privacy, and security
- quality checks and versioning
- retention, sharing, and reproducibility

Ethical and data constraints can change the sample, measurement, and scientific claim.

Ethics and data governance belong inside the method logic, not in a detached compliance paragraph

Risk Needs Triggers and Fallbacks

Risk	Trigger	Mitigation	Fallback	Owner
External data delayed	No agreement by month 3	Start governance in month 1	Temporal validation only; narrow claim	PI + site lead
Too few outcome events	Below event minimum at month 4	Monitor blinded counts	Reduce model complexity; revise scope	Statistician

A contingency plan is credible when it states when action begins, who acts, and how the claim changes

Exercise 3 · Build the Next Three Links

GIVEN GAP

Whether a locked EHR model retains discrimination and calibration over time and at another hospital is uncertain.

GIVEN QUESTION

How do its discrimination and calibration compare with the prespecified baseline temporally, and are they retained externally?

OBJECTIVE	METHOD	CONTRIBUTION
What knowledge will the project produce?	What design and analysis can produce credible evidence?	What conclusion is justified, and what remains unproven?

GROUP TASK

Complete all three boxes; limit claims to predictive performance.

The research question constrains the objective, method, and permissible contribution

Sample Answer

OBJECTIVE

Estimate and compare the discrimination and calibration of the locked model and prespecified baseline in later-period and external cohorts.

METHOD

Freeze both models after development; use prespecified cohorts and outcomes; evaluate later-period and external cohorts without retuning; report metrics, differences, and confidence intervals.

CONTRIBUTION

Quantify comparative and transport performance under defined conditions. The study does not establish clinical benefit or workflow impact.

A good answer states both the intended contribution and the limit of the claim

Exercise 4 · Transfer to a New Topic

TEACHING CASE

Generative-AI feedback for novice programmers

ILLUSTRATIVE EVIDENCE PACK

- course goal: debugging performance without AI assistance
- AI assistance may improve completion on the current task
- some AI-generated suggestions contain errors
- performance after assistance is removed remains uncertain
- transfer to unseen debugging tasks is not established in the case materials

GROUP TASK

Write:

1. one NEED sentence

2. one bounded GAP

3. one research QUESTION

4. one essential design feature

Do not claim that AI improves education. Define an outcome that demonstrates independent transfer.

Transfer the reasoning structure, not the wording of the previous example

Sample Answer -- Test Transfer, Not Only Immediate Performance

NEED	Novice programmers need feedback that supports independent debugging, not only completion of the current exercise.
GAP	Immediate task performance does not show whether students retain the skill after AI assistance is removed or transfer it to unseen debugging problems.
QUESTION	Among novice programmers, does structured generative-AI feedback improve performance on an unseen debugging task one week later compared with standard hints?
DESIGN	Randomize after a common pretest; use an immediate task, a delayed unseen task, blinded scoring, and an audit of inaccurate AI suggestions.

Contribution: evidence about delayed transfer and feedback risks in this course setting—not a general claim about AI and education.

The outcome and timing must match the claim about independent learning



To be continued